HERMES · ARCHITECTURE & SESSION BRIEF

Memory System: Decisions & Overhaul

v2-only · BGE-large · shared skills · per-agent visibility

Merged brief combining the architectural-decisions deck with the one-day overhaul summary: what we kept, what we cut, why the embedder finally got fixed, the orphan cron we found, the plugin patches we shipped, the tooling we built — and what to read next.

2026-05-17 · OpenClaw · *Prepared for technical co-founder review*

Sources merged: docs/architecture/2026-05-17-memory-system-decisions.pptx + docs/architecture/2026-05-17-session-summary.pptx

TL;DR · FIVE WINS · ONE ROOT CAUSE

Where we landed after one day.

v2 plugin chosen. Auto-promoter killed. Patches durable. Stress test green.

01	**Committed to v2 plugin**	v1 MemOS server retired — no Qdrant/Neo4j to babysit. Single SQLite, in-process, row-level tenancy.
02	**BGE-large embeddings**	1024-dim, local ONNX. +14% MRR over MiniLM, beats gemini-embedding-2 on every quality metric. $0/mo.
03	**Auto-promoter killed**	Orphan Sprint-3 cron writing share_scope=local every 15 min — disabled, with forensic triggers as a tripwire.
04	**Per-agent viewer toggle**	AsyncLocalStorage + X-As-Profile header. One daemon, any agent's lens — no second process.
05	**Open-world speaker attribution**	Names from [handles], self-intros, quoted bylines. The MEMOS_HUMANS list is help, not a gate.

FINAL STATE 36 / 36 stress-test checks green · 13 / 13 plugin patches verified · durable via postinstall-patches.sh

PART 1

Architectural Decisions

One machine · one plugin · row-level tenancy

CONTEXT

Three things were being conflated

Operator confusion was warranted — three memory ideas with similar names, only one actually doing work.

SYSTEM 1

MemOS v1 server

IDLE

Standalone HTTP service on :8001. Qdrant + Neo4j + SQLite. Cube-based isolation via API-key → user_id binding. The original production target.

SYSTEM 2

v2 memos plugin

ACTIVE

Hermes-native plugin. Single SQLite + UI on :18800. Reward backprop, L1/L2/L3 layers, automatic skill crystallisation. Isolation by row-level namespace tuple.

SYSTEM 3

The hub

PLANNED

Future cross-machine sync. Config schema and share_scope='hub' exist; zero code reads them. The failing systemd unit was trying to start it.

ARCHITECTURE

What's actually running

One system is doing the work. Two are inert. One was thrashing.

COMPONENT	ROLE	STATUS	LOCATION
v2 memos-local-plugin	Sole active memory provider	ACTIVE	`~/.hermes/memos-plugin/ ::18800`
MemOS v1 server	HTTP API; ceo-cube empty (zero writes)	STOPPED	`localhost:8001 (disabled)`
memos-toolset	v1 client; plugin.yaml.disabled-2026-05-12	INERT	`~/.hermes/plugins/memos-toolset/`
memos-hub.service	Tries to start non-existent hub mode	DISABLED	`scripts/migration/bootstrap-hub.sh`
Hermes agent gateways	Per-profile processes (each writes to v2)	ACTIVE	`hermes-gateway-{sergio,hr-agent,…}`

→ ***Net: zero actual contamination, ever. v1 cube was empty when stopped — probed directly: text_mem=[], total_nodes=0.***

DECISION · BACKEND

v2 plugin chosen. v1 retired.

Paperclip + CEO orchestration retired with it — single machine, single tier.

ASPECT	v1 MemOS server (retired)	v2 @memtensor/memos-local-plugin (kept)
Stores	Qdrant + Neo4j + SQLite	Single SQLite + sqlite-vss
Isolation	Per-cube API keys + URL	Row-level (owner_agent_kind, owner_profile_id, share_scope)
Sharing	Cross-cube replication (untested)	share_scope=local rows visible across profiles, in-process
Footprint	3 services + REST API	1 node daemon, embeds Hermes-side
At session start	Cube empty — nothing to migrate	All agents already writing through it
UI	None	Bundled memory browser + 2D/3D UMAP viewers on :18800
Decision	Stop + disable systemd unit	**Patch in place, ship**

WHY THE REVERSAL VS THE APRIL v2-DEPRECATED AUDIT?

The April audit flagged v2 because it lacked operational tooling. We added that tooling (cross-agent inspector, forensic-audit triggers, per-agent viewer, stress test) instead of running a second backend. Cost / observability now both better than v1.

PRIMITIVE

How v2 isolation actually works

Not cubes. Row-level multi-tenancy with a namespace tuple stamped on every row.

core/runtime/namespace.ts

```
// every row carries these columns
owner_agent_kind   TEXT NOT NULL
owner_profile_id   TEXT NOT NULL
share_scope        TEXT DEFAULT 'private'

// every read is filtered through this
function isVisibleTo(row, caller) {
  if (row.share_scope !== 'private') return true;
  return (
    row.owner_agent_kind === caller.agentKind &&
    row.owner_profile_id === caller.profileId
  );
}
```

THE PRIMITIVE

`(agentKind, profileId, workspaceId)`

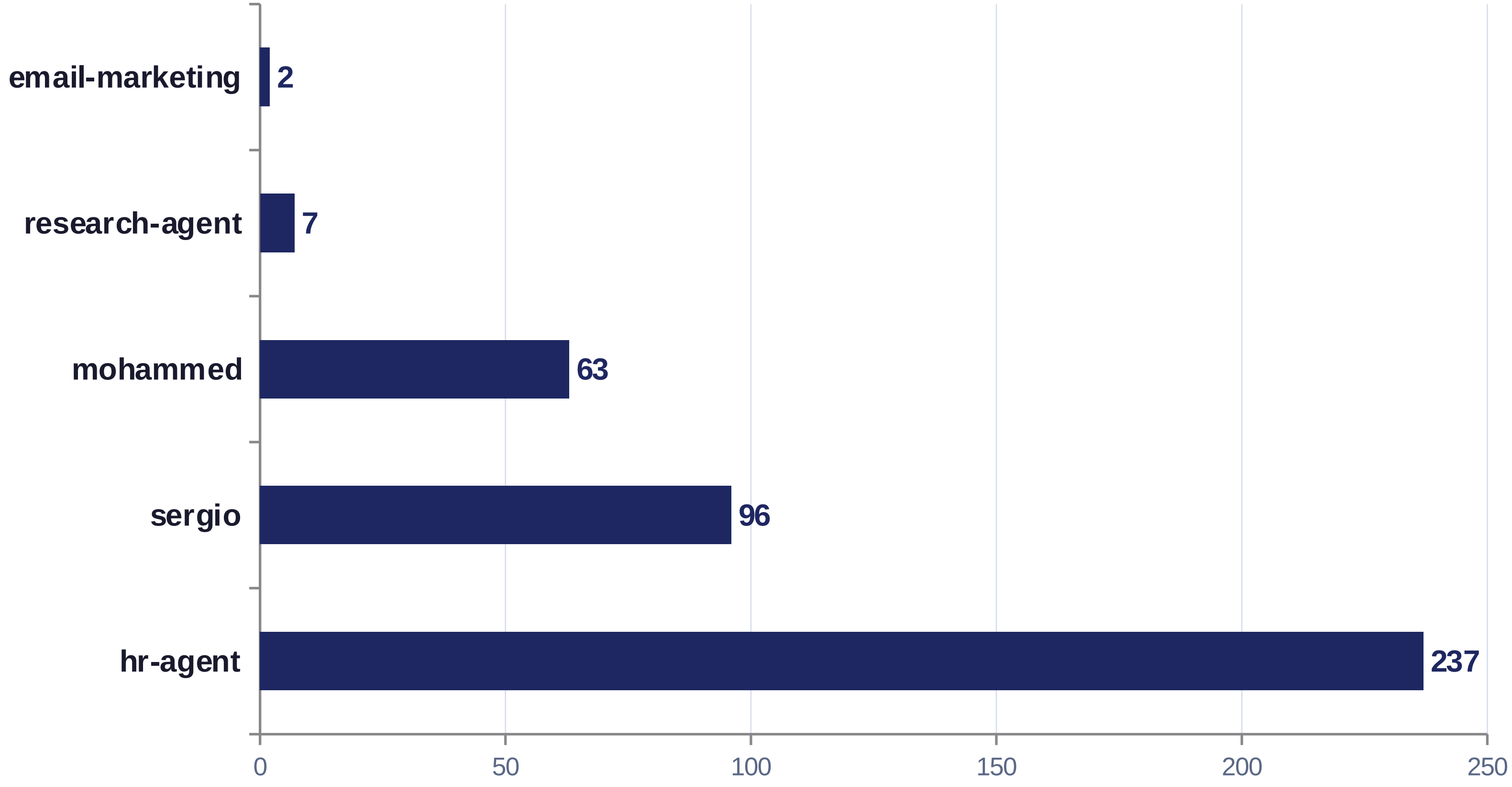
WHY IT WORKS

- One SQLite, many tenants — no cubes, no per-agent files
- Enforced at every retrieval, not by trust
- No server, no API, no credential management
- Plugin derives profileId from $HERMES_HOME — each agent process gets its own namespace automatically

EVIDENCE

Isolation is working — verified on the live DB

SELECT owner_profile_id, COUNT(*) FROM traces GROUP BY owner_profile_id



VERDICT

No "default" bucket.

Every agent's writes land under its own **profileId**.

The earlier cross-contamination worry was real for v1's shared API key — but **moot for v2**, which derives profileId from $HERMES_HOME at process start.

SHARING POLICY

Private experience. Shared knowledge.

Raw turn data stays with the agent who lived it. Crystallised skills + world facts go to the team.

TABLE	SCOPE	EFFECT	RATIONALE
`traces`	`private`	Each agent sees only its own turns	*Raw conversation history is personal — no leak across agents*
`episodes`	`private`	Session boundaries stay per-agent	*Episode keys are agent-local context, not shared schema*
`policies`	`private`	L2 candidate patterns are local experiments	*Half-baked heuristics shouldn't pollute the team's policy space*
`world_model`	`local`	Environment + operator facts shared across agents	*"Sergio prefers MiniMax for synthesis" should help every agent*
`skills`	`local`	Crystallised callable procedures shared across agents	*Skills are the team's collective know-how — attribution via (learned by <profile>)*

isVisibleTo RULE · core/runtime/namespace.ts:136

A row is visible to a caller IFF `share_scope ∈ {local, public, hub}` OR `(owner_agent_kind, owner_profile_id)` matches the caller's namespace. Per-row attribution is automatic.

share_scope: design intent vs reality today

Tag with the intended scope anyway — when the hub ships, the data already knows where it should go.

SCOPE	DESIGN INTENT	TODAY'S BEHAVIOR	AS DESIGNED
`private`	Visible only to the writing profile	Only one that actually filters	✓
`local`	Visible across profiles on this machine	Visible to all (same as public)	✗
`public`	Visible to everyone, everywhere	Visible to all	✓
`hub`	Visible across machines via hub sync	Visible to all — hub never wired	✗

Implementation collapses three of four scopes into one today. That's intentional — single-machine deployment means private vs not-private is the only meaningful split until the hub ships. The schema is ready for when it does.

PART 2

Embedder Overhaul

200 traces · 8 queries · DeepSeek V3 as judge · BGE-large wins everything

METHODOLOGY

Embedder benchmark: how we tested

Three models, eight realistic queries, LLM-graded relevance on the union of top-5 from every model.

01

Pull real traces

200 trace summaries from the live v2 DB (read-only). Mixed profiles, real content — embedder choice, file transfer, Idealista, plugin architecture.

02

Embed with all three

MiniLM-L6-v2 (current), BGE-large-en-v1.5 (proposed local), gemini-embedding-2 (proposed cloud). BGE used recommended query prefix.

03

Query set + retrieval

8 realistic memory queries. For each: top-5 retrieval from every model. Cosine on L2-normalised vectors.

04

LLM-as-judge

DeepSeek V3 rated each (query, snippet) pair on 0/1/2 relevance. Judged the UNION of top-5 across all models — no free passes.

METRICS Precision@5 · MRR · nDCG@5 · pairwise top-5 overlap (Jaccard) · top-1 cosine similarity

RESULTS

3-way benchmark: BGE-large wins everything

P@5 · MRR · nDCG@5 — top of every metric, on a benchmark drawn from real Hermes traces.

MODEL	DIM	P@5	MRR	nDCG@5	RUNTIME	COST / MO	VERDICT
`Xenova/all-MiniLM-L6-v2`	384	**0.450**	**0.688**	**0.628**	Local (Xenova)	$0	baseline
`Xenova/bge-large-en-v1.5`	1024	**0.500**	**0.781**	**0.725**	Local (Xenova)	$0	WINNER
`Google gemini-embedding-2`	1536	**0.425**	**0.750**	**0.662**	Google API	≈ $14	STRONG

Δ vs MiniLM

+11% P@5 · +14% MRR · +15% nDCG@5

Δ vs Gemini-2

+18% P@5 · +4% MRR · +10% nDCG@5

Sample

200 traces · 8 queries · DeepSeek-judged

RATIONALE

Why BGE-large is the pick

Top quality · stays local · no API · acceptable resource cost.

QUALITY

Top of every metric

P@5, MRR, nDCG@5 — all three. Q4 (Idealista cluster) and Q7 (skill crystallisation) found content the others missed entirely.

LOCAL

No API dependency

Runs in the Hermes process via Xenova/Transformers.js ONNX. No network round-trips, no rate limits, no Google account to manage.

PRIVACY

Memory never leaves the box

Every other option meant agent memory content was sent to a third-party API. BGE-large keeps it on the machine where it belongs.

FOOTPRINT

Acceptable cost

~1.5 GB RAM working set; machine has ~9 GB free. ~150–300 ms per embedding on CPU. The hook is async — doesn't block agent replies.

MODEL ASSIGNMENTS

Three models, three jobs.

Picked for fit, not flag-planting — extraction is cheap, crystallisation is rare, embeddings are hot path.

Capture / extraction

DeepSeek V3

`OpenAI-compatible · deepseek-chat`

Non-thinking model — no <think> pollution in summaries. Cheap enough per-turn to never think about. Strict-JSON path with heuristic fallback if malformed.

Skill crystallisation

MiniMax M2.7

`Anthropic-compatible · api.minimax.io/anthropic`

Reasoning-heavy synthesis from multi-turn evidence. Rare call — quality > latency. Output post-processed; <think> tags stripped cleanly.

Embedding

Xenova/bge-large-en-v1.5

`In-process · ONNX · 1024-dim`

Best MRR in our bench. Local — no API for a hot-path call, no privacy footprint, no rate limits.

PART 3

The Orphan Cron

Every 15 minutes, share_scope went back to local. Symptom looked like the plugin — it wasn't.

The auto-promoter root cause

Manual demotion held for 14 minutes, then bulk-reverted. The plugin wasn't doing it.

1 Symptom

Manually flipped traces to private. Within 15 min, all rows back to local. Reproducible.

2 Forensic SQLite triggers

tools/forensic-audit.sql installs no-op AFTER-UPDATE triggers logging share_scope writes to a side table. Non-modifying — safe.

3 Next cycle, caught the actor

Bulk UPDATE pattern with no plugin sql_text — written directly via the SQLite CLI. Plugin's HTTP API + namespace filter bypassed entirely.

4 Traced to a cron

crontab -l revealed a */15 entry running python3 on …/promote-memos-shares.py inside an orphan Claude worktree (.claude/worktrees/nice-mclaren-13f017/scripts/).

5 It was Paperclip/CEO scaffolding

Sprint 3 (2026-05-12) added it so the orchestrator could read across cubes. We retired Paperclip 5 days later — the cron lived on.

6 Disabled, not deleted

Commented out in crontab with rationale. Forensic triggers stay installed as a tripwire for re-emergence.

PART 4

Plugin Patches

13 files · 5 logical changes · no upstream repo · re-applied via postinstall hook

13 files patched, applied via postinstall hook.

@memtensor/memos-local-plugin@2.0.0 ships with no public repo — patches live in tools/plugin-patches/ and re-apply on every npm install.

#	LOGICAL CHANGE	FILES	WHAT IT DOES
1	**Shared-skill attribution**	4	renderSkill appends '(learned by <profile>)' so agents see which profile crystallised a shared skill.
2	**Per-request namespace override**	4	AsyncLocalStorage context parsed from X-As-Profile / ?as_profile=, applied at SQL time in trace pagination.
3	**Bundled-viewer UI overlay**	2	Floating 'view as <profile>' picker injected into web/dist — intercepts fetch() to add the header.
4	**Skill packager default scope**	1	New skills default share_scope=local (was private) — no more manual promotion after crystallisation.
5	**Named-speaker memory summaries**	2	Summarizer reads MEMOS_HUMANS env, attributes speech to actual names — open-world (handles + self-intros).

Run: `bash tools/postinstall-patches.sh` **Verify:** `bash tools/postinstall-patches.sh --check`

PATCH SPOTLIGHT · #2 + #3

Per-agent viewer toggle.

One daemon, one DB. Any agent's perspective on demand.

SERVER · ALS namespace

core/runtime/request-namespace.ts (new)

```
const als = new AsyncLocalStorage<RuntimeNamespace>();

export function runWithRequestNamespace(ns, fn) {
  return als.run(ns, fn);
}
export function getRequestNamespace() {
  return als.getStore();
}
```

CLIENT · fetch interceptor

web/dist/hermes-profile-switcher.js (new)

```
const orig = window.fetch.bind(window);
window.fetch = (input, init) => {
  if (!currentProfile) return orig(input, init);
  const headers = new Headers(init?.headers);
  headers.set('X-As-Profile', currentProfile);
  return orig(input, { ...init, headers });
};
```

`server/http.ts` wraps dispatch() in `runWithRequestNamespace(ns, …)`. `memory-core.ts` effectiveNamespace() prefers ALS over the daemon's startup namespace. `traces.ts` applies it at SQL time so paginated turn-keys actually match the requested profile.
Active at **https://tower.taila4a33f.ts.net/** — top-right dropdown.

PART 5

Tooling We Built

Inspectors · forensic triggers · stress test · 2D + 3D embedding-space viewers

CROSS-AGENT TOOLING

Inspect anything. Audit everything.

Each tool lives in tools/. Run from anywhere on the box — no extra setup.

TOOL	PURPOSE	COMMON INVOCATION
`tools/memos-explorer.py`	Cross-agent CLI: list profiles, traces, skills, policies, world facts. Vector-space stats, similarity search, project per profile.	`python3.12 tools/memos-explorer.py profiles` `python3.12 tools/memos-explorer.py search 'auth' --top 5`

`tools/forensic-audit.sql`	Installs AFTER-UPDATE/INSERT/DELETE triggers logging every share_scope write to share_scope_audit. Non-modifying tripwire.	`sqlite3 ~/.hermes/memos-plugin/data/memos.db \`
		`  < tools/forensic-audit.sql`
		`python3.12 tools/memos-explorer.py audit`

`tools/stress-test.sh`	36 end-to-end checks: patch presence, daemon health, share-scope policy, per-agent visibility, embedding dims, capture pipeline.	`bash tools/stress-test.sh` `# expect: 36 passed, 0 failed`
`tools/postinstall-patches.sh`	Idempotent applier of all 13 plugin patches + symlinks for UMAP / graph viewers. Run after every npm install of the plugin.	`bash tools/postinstall-patches.sh` `bash tools/postinstall-patches.sh --check`

2D Plotly + 3D deck.gl.

Sanity-check what the BGE-large embedder actually believes — interactively.

2D · PLOTLY

`tools/umap-viewer.html`

Static UMAP projection. Color by profile, hover shows summary, click opens full trace in the side panel.

URL: `https://tower.taila4a33f.ts.net/umap-viewer.html`

Reads: `tools/vec-map.json`

BEST FOR

A quick read on cluster topology, identifying outliers.

3D · DECK.GL · GPU

`tools/umap-3d-viewer.html`

Orbit camera, GPU-rendered point cloud, drag-to-rotate. Cross-agent rows wear white halos so they pop against any profile color.

URL: `https://tower.taila4a33f.ts.net/umap-3d-viewer.html`

Reads: `tools/vec-map-3d.json`

BEST FOR

Spotting bridges between profiles, seeing structure the 2D projection flattened.

UX TOUCHES

Profile palette redesigned so no agent color collides with the ember accent · cross-agent indicator switched from orange tint to white halo · persistent plain-language "How to read this map" panel.

PART 6

Speaker Attribution

Names come from signal, not from a roster. Handles · self-intros · quoted bylines.

PATCH SPOTLIGHT · #5

Names from signal, not from a roster.

Agents see messages from outside parties too — clients, leads, group chats, quoted email. The list is help, not a gate.

MEMOS_HUMANS (in ~/.hermes/.env)

```
MEMOS_HUMANS=Sergio:sergiopalacio96|sergiop,Krati,Mohammed:moh,Arinze
```

Format: Name[:handle1|handle2|...], ... Name-only entries are fine — they still help when the body mentions the name.

SIGNAL	EXAMPLE	RESULTING SUMMARY
Gateway-prepended handle	`[sergiopalacio96] can you check the cron?`	*Sergio asked to verify the cron schedule.*
Self-introduction in body	`Hi, I'm Mark from acme.io — want a quote on …`	*Mark from acme.io requested a quote.*
Quoted email From: header	`From: anna@bizpartner.co\nWe'd like to renew …`	*Anna requested a renewal.*
No identifiable speaker	`(direct message body with no marker)`	*Speaker reference omitted — never invented.*

Files: `core/capture/summarizer.ts` · `bridge.cts` *(dotenv loader so the daemon sees ~/.hermes/.env on its own)*

PART 7

Final State

Cleanups · config.yaml · migration steps · verification

SUBTRACTION

Cleanups: what we removed

Each removal is reversible — none are destructive deletes. Service stops and config swaps only.

DISABLED	STOPPED	RETIRED	REPLACED	DISABLED
`memos-hub.service`	`memos-server.service (v1)`	`Paperclip / CEO orchestration`	`gemini-embedding-2 config`	`promote-memos-shares.py cron`
Was thrashing in an auto-restart loop, calling bootstrap-hub.sh which tries to launch a hub mode the plugin doesn't implement. Zero functional impact.	Healthy but unused — cube was empty, memos-toolset client already disabled. Frees Qdrant + Neo4j + Python memory. Restart anytime if we revisit v1.	Single-machine architecture means no CEO routing. Skill patches authored directly. Removed 2026-05-17 along with multi-cube assumptions.	Briefly swapped in during the embedder eval. Beaten by BGE-large on every quality metric. Reverted to local embedder before any data was written.	Sprint-3 orphan that bulk-updated share_scope every 15 min, bypassing the plugin's API. Commented out in crontab with rationale. Triggers stay installed as a tripwire.

FINAL STATE · ~/.hermes/memos-plugin/config.yaml

Where we landed.

One local embedder, one cheap extractor, one reasoning model. Hub flag is cosmetic.

```yaml
embedding:
  provider:    local
  model:       Xenova/bge-large-en-v1.5     # +11–18% over MiniLM / Gemini-2
  dimensions:  1024

llm:
  provider:    openai_compatible
  endpoint:    https://api.deepseek.com
  model:       deepseek-chat                  # extraction / summarisation

skillEvolver:
  provider:    anthropic
  endpoint:    https://api.minimax.io/anthropic
  model:       MiniMax-M2.7                   # crystallisation / L2 induction

hub:
  enabled:     true                                # cosmetic — feature unimplemented upstream
```

Isolation is per-profile via $HERMES_HOME → profileId. No cubes, no shared credentials. Verified on the running DB.

Migration steps remaining

A clean one-minute downtime; no recoverable data is lost (the 30 skills were all status=candidate).

01

Stop Hermes agents

```
systemctl --user stop hermes-gateway.service hermes-gateway-*
```

02

Wipe v2 DB (384d → 1024d incompatible)

```
rm -f ~/.hermes/memos-plugin/data/memos.db*
```

03

Restart agents — DB and embeddings rebuild via BGE

```
systemctl --user start hermes-gateway.service hermes-gateway-*
```

04

Tail one agent — verify BGE model loads cleanly

```
journalctl --user -u hermes-gateway-sergio -f
```

VERIFICATION

All green. Durable.

The patches re-apply themselves on every npm install. Drift gets caught by audit triggers.

36 / 36	**13 / 13**	**1 / 1**	**0**
STRESS-TEST CHECKS	PLUGIN PATCHES VERIFIED	AUTO-PROMOTER KILLED	POSTINSTALL STEPS

KEEP IT GREEN

WHEN	RUN
After every npm install/update of @memtensor/memos-local-plugin	`bash tools/postinstall-patches.sh`
After any change to core daemon code / share policy	`bash tools/stress-test.sh`
If share_scope ever drifts unexpectedly again	`python3.12 tools/memos-explorer.py audit`

OPEN FOLLOW-UPS

Not blocking. Worth doing.

Three lines we picked up but didn't pull all the way through.

Cluster summarisation in the UMAP viewers

Ask DeepSeek V3 to one-line each HDBSCAN cluster (≈31 calls). Render labels in 2D + 3D viewers so the topology becomes self-explanatory.

`tools/umap-viewer.html` · `tools/umap-3d-viewer.html`

Per-profile MEMOS_HUMANS

Today MEMOS_HUMANS is a single env var. Expand to MEMOS_HUMANS__<profile> so each agent can carry its own alias map (e.g., research-agent knows its lead list).

`core/capture/summarizer.ts`

Backfill historical traces with the named-speaker prompt

Existing summaries still say "the user". A re-summarisation pass against the new prompt would put names into older embeddings and align them with new ones.

`tools/memos-explorer.py (new resummarize subcommand)`

POINTERS

Where to read the rest.

Everything the next agent (human or otherwise) needs to come up to speed.

DOCUMENT	WHAT'S IN IT
`CLAUDE.md (this repo, root)`	Active sprint header + the v2-only memory architecture rules every new agent reads first.
`memos-setup/learnings/2026-05-17-v2-only-bge-shares.md`	Canonical decision doc — backend choice, embedder benchmark, share policy, rollback plan.
`docs/architecture/2026-05-17-memory-system-decisions.pptx`	Original architecture deck — preserved as one of the two sources of this merge.
`docs/architecture/2026-05-17-session-summary.pptx`	Original session-summary deck — the other source of this merge.
`tools/plugin-patches/`	1:1 mirror of every patched file — the canonical "what changed in v2".
`tools/forensic-audit.sql ( + .down.sql )`	Non-modifying SQLite triggers — install once, audit forever.
`https://tower.taila4a33f.ts.net/`	Memory viewer (password-gated). Top-right dropdown switches the lens to any agent.

TRY IT

Five things to do tomorrow.

Each one takes under two minutes.

1

Open the 3D viewer

Orbit. Switch profile. White halos = rows visible across multiple agents.

```
xdg-open https://tower.taila4a33f.ts.net/umap-3d-viewer.html
```

2

Send a message to an agent — watch summary

Use the [handle] convention if you want named attribution.

```
hermes send sergio '[sergiopalacio96] what is the embedder cost again?'
```

3

Audit recent share_scope writes

If anything writes share_scope directly to SQLite, you'll see it here.

```
python3.12 tools/memos-explorer.py audit
```

4

Run the full stress test

36 checks. Should print 36 passed, 0 failed in well under a minute.

```
bash tools/stress-test.sh
```

5

Use the per-agent toggle in the viewer

Top-right dropdown. Switching profile re-fetches every list with X-As-Profile.

```
visit https://tower.taila4a33f.ts.net/ and pick a profile
```

END OF BRIEF

One day. One memory system.

v2 plugin chosen · BGE-large wired · auto-promoter killed · per-agent viewer shipped · open-world speaker attribution.

Questions worth asking next session

- How do skills propagate when a new profile spins up — instant via shared layer, or does it need a recall pass?
- Should cluster labels in the UMAP viewers be cached, or recomputed every embedding-set rebuild?
- What's the right TTL on a private trace before it's eligible for archival to a cold layer?
- When does the hub become more than a schema reservation — what's the trigger event?